

Lab Manual Route 53

Steps

1. Prepare the EC2 instance with webserver installed on it
2. Register a domain name with Route 53
3. To Create DNS record

Step1: Prepare the EC2 instance with webserver installed on it

Step 1: Choose an Amazon Machine Image (AMI)

[Cancel and Exit](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. You can select an AMI provided by AWS, our user community, or the AWS Marketplace, or you can select one of your own AMIs.

Search for an AMI by entering a search term e.g. "Windows"

Quick Start

My AMIs

AWS Marketplace

Amazon Linux 2 AMI (HVM), SSD Volume Type - ami-0b69ea66ff7391e80 (64-bit x86) / ami-09c61c4850b7465cb (64-bit Arm)

Free tier eligible

Amazon Linux 2 comes with five years support. It provides Linux kernel 4.14 tuned for optimal performance on Amazon EC2 systems.

Select

64-bit (x86)

Filter by:

All instance types

Current generation

Show/Hide Columns

Currently selected: t2.micro (Variable ECUs, 1 vCPUs, 2.5 GHz, Intel Xeon Family, 1 GiB memory, EBS only)

	Family	Type	vCPUs	Memory (GiB)	Instance Storage (GB)	EBS-Optimized Available	Network Performance	IPv6 Support
☐	General purpose	t2.nano	1	0.5	EBS only	-	Low to Moderate	Yes
☒	General purpose	t2.micro Free tier eligible	1	1	EBS only	-	Low to Moderate	Yes

Step 3: Configure Instance Details

Configure the instance to suit your requirements. You can launch multiple instances from the same AMI, request Spot instances to take advantage of the lower pricing, assign an access management role to the instance, and more.

Number of instances

1

Launch into Auto Scaling Group

Purchasing option

☐ Request Spot instances

Network

vpc-0b17f9a14de147cc1 (default)

Create new VPC

Subnet

No preference (default subnet in any Availability Zone)

Create new subnet

Auto-assign Public IP

Use subnet setting (Enable)

Placement group

☐ Add instance to placement group

T2/T3 Unlimited

☐ Enable

Additional charges may apply

Advanced Details

User data

☒ As text
 ☐ As file
 ☐ Input is already base64 encoded

```
#!/bin/bash
sudo yum install httpd -y
service httpd start
chkconfig httpd on
touch /var/www/html/index.html
echo "THIS IS A TEST WEBSERVER" >> /var/www/html/index.html
```

Cancel

Previous

Review and Launch

Next: Add Storage

AWS – Route 53

Step 4: Add Storage

Your instance will be launched with the following storage device settings. You can attach additional EBS volumes and instance store volumes to your instance, or edit the settings of the root volume. You can also attach additional EBS volumes after launching an instance, but not instance store volumes. [Learn more](#) about storage options in Amazon EC2.

Volume Type ⓘ	Device ⓘ	Snapshot ⓘ	Size (GiB) ⓘ	Volume Type ⓘ	IOPS ⓘ	Throughput (MB/s) ⓘ	Delete on Termination ⓘ	Encryption ⓘ
Root	/dev/xvda	snap-0e4c15b8cba3e8ae6	8	General Purpose SSD (gp2)	100 / 3000	N/A	☒	Not Encrypt
Add New Volume								

Step 6: Configure Security Group

A security group is a set of firewall rules that control the traffic for your instance. On this page, you can add rules to allow specific traffic to reach your instance. For example, if you want to set up a web server and allow Internet traffic to reach your instance, add rules that allow unrestricted access to the HTTP and HTTPS ports. You can create a new security group or select from an existing one below. [Learn more](#) about Amazon EC2 security groups.

Assign a security group: ☐ Create a new security group
☒ Select an existing security group

Security Group ID	Name	Description	Actions
☐ sg-0d2b3ee176e5227b4	default	default VPC security group	Copy to new
☒ sg-073d06602f3450560	SSH-HTTP	SSH-HTTP	Copy to new

Inbound rules for sg-073d06602f3450560 (Selected security groups: sg-073d06602f3450560)

Type ⓘ	Protocol ⓘ	Port Range ⓘ	Source ⓘ	Description ⓘ
HTTP	TCP	80	0.0.0.0/0	
HTTP	TCP	80	:::0	
SSH	TCP	22	0.0.0.0/0	
SSH	TCP	22	:::0	

[Cancel](#) [Previous](#) [Review and Launch](#)

Filter by tags and attributes or search by keyword

1 to 1 of 1

☒	Name	Instance ID	Instance Type	Availability Zone	Instance State	Status Checks	Alarm Status	Public DNS (IPv4)
☒		i-017603b34d39d0f02	t2.micro	us-east-1a	● running	Initializing	None	ec2-54-164-27-87.com

Instance: i-017603b34d39d0f02

Public DNS: ec2-54-164-27-87.compute-1.amazonaws.com

Description

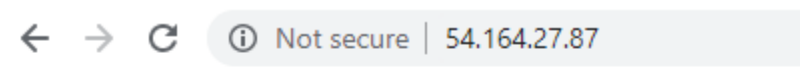
Status Checks

Monitoring

Tags

Instance ID	i-017603b34d39d0f02	Public DNS (IPv4)	ec2-54-164-27-87.compute-1.amazonaws.com
Instance state	running	IPv4 Public IP	54.164.27.87
Instance type	t2.micro	IPv6 IPs	-

Output of the webserver



THIS IS A TEST WEBSERVER

Step2: Register a domain name with Route 53

Note: -- This is charged around 12\$ per year

Registered domains

[Register Domain](#) [Transfer Domain](#) [Domain Billing Report](#)

Domain Name	Privacy Protection	Expiration Date	Auto
vishwacloudlab.net	All contacts	July 01, 2020	✓

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[Hosted zones](#)
[Health checks](#)
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[Traffic policies](#)
[Policy records](#)
[Domains](#)
[Registered domains](#)
[Pending requests](#)
[Resolver](#)
[VPCs](#)

[Back to Hosted Zones](#) [Create Record Set](#) [Import Zone File](#) [Delete Record Set](#) [Test Record Set](#)

☐ Aliases Only ☐ Weighted Only

Displaying 1 to 2 out of 2 Record Sets

Value	Evaluate Target Health	Health Check ID	TTL	Reg
ns-1998.awsdns-57.co.uk.	-	-	172800	
ns-1522.awsdns-62.org.	-	-		
ns-51.awsdns-06.com.	-	-		
ns-978.awsdns-58.net.	-	-		
ns-1998.awsdns-57.co.uk. awsdns-hostmaster.amaz	-	-	900	

To get started, click Create Record Set button or click an existing record set.

Step3:

To Create a DNS record .

The screenshot shows the AWS Route 53 console interface. At the top, the 'Create Record Set' button is highlighted with a red circle. Below this, a table displays existing records for the 'vishwacloudlab.net' zone. The table has columns for Name, Type, Value, and Evaluation. Two records are listed: an NS record and an SOA record. Below the table, the 'Create Record Set' form is shown. It includes fields for Name (ws01.vishwacloudlab.net), Type (A – IPv4 address), Alias (No), TTL (300 seconds), Value (54.164.27.87), and Routing Policy (Simple). A 'Create' button is at the bottom of the form.

Name	Type	Value	Evaluation
vishwacloudlab.net	NS	ns-1998.awsdns-57.co.uk. ns-1522.awsdns-62.org. ns-51.awsdns-06.com. ns-978.awsdns-58.net.	-
vishwacloudlab.net	SOA	ns-1998.awsdns-57.co.uk. awsdns-hostmaster.amaz	-

Create Record Set

Name: ws01.vishwacloudlab.net

Type: A – IPv4 address

Alias: ☐ Yes ☒ No

TTL (Seconds): 300 1m 5m 1h 1d

Value: 54.164.27.87

IPv4 address. Enter multiple addresses on separate lines.
Example:
192.0.2.235
198.51.100.234

Routing Policy: Simple

Route 53 responds to queries based only on the values in this record. [Learn More](#)

Create

Output.

